

AULYN -- CS201: Recursion & Call Stack Guide

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1. Key Elements of Recursion:

2. Call Stack Tracing:

- Every recursive invocation pushes a new stack frame containing local variables and return address onto the call stack.
- Stack overflow occurs if no base case is hit before memory limits.

3. Example: Factorial $n! = n * (n-1)!$

- Base case: if $n \leq 1$ return 1